SMART CAMPUS
LOST AND FOUND SYSTEM

A Java Console Application to Report, Search, Match and Return Lost and Found Items on a Campus

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice:
```

Technology	Java (JDK 8 or above)
Interface	Console menu (Scanner)
Data Storage	Text file (data/items.txt)
Source File	LostAndFoundSystem.java

1. Introduction

On a busy campus, students lose phones, books, ID cards and wallets every day, and the people who find them often do not know whom to give them to. Notice boards and chat groups are slow and are not searchable. The **Smart Campus Lost and Found System** keeps all lost and found reports in one place.

Anyone can report a lost or found item, list and search the reports, and mark an item as returned. The *smart* part is automatic matching: whenever a new report is entered, the system compares it with the open reports of the opposite type (lost against found) and displays the likely matches with a match percentage, together with the contact details needed to verify the owner.

2. Objectives

- Collect lost and found reports with item name, category, place, date, description and contact details.
- Match lost items with found items automatically using a simple scoring rule.
- List the reports with filters (all, lost, found, open) and search them by keyword.
- Close both related reports when an item is returned to its owner.
- Show statistics such as the recovery rate, reports per category and the busiest location.
- Validate every input (date format, 10-digit phone number, valid category, no empty fields).
- Save the reports permanently in a file.

3. Modules of the System

Menu Option	Module	Description
1	Report Lost Item	Reads the details of a lost item, saves it and shows possible matches.
2	Report Found Item	Reads the details of a found item, saves it and shows possible matches.
3	View Items	Lists reports using one of four filters: all, lost only, found only, open only.
4	Search Items	Finds reports by ID, item name, category, location or description keyword.
5	Smart Match	Shows the found items that match a given lost item, with the match percentage.
6	Mark Item as Returned	Closes an item after the owner is verified, and optionally its related report.
7	Statistics Report	Shows counts, recovery rate, reports by category and the busiest location.
0	Exit	Ends the program.

4. Tools and Technologies

Item	Details
Language	Java (JDK 8 or above; tested with OpenJDK 21)
Input / Output	java.util.Scanner and System.out (console)
Date handling	java.time.LocalDate and ChronoUnit (date validation and day differences)
Collections	ArrayList for the reports, TreeMap and HashSet for statistics and matching
File Handling	java.io (BufferedReader, FileReader, FileWriter, PrintWriter)
Data Storage	Plain text file data/items.txt, fields separated by the | character

5. System Design

5.1 Item class

A single model class, **Item**, represents both lost and found reports; the *type* field tells them apart.

Member	Type	Description
id	int	Unique report ID, assigned automatically (highest existing ID + 1)
type	String	LOST or FOUND
name	String	Item name, for example Samsung Phone
category	String	Electronics, Books, ID Cards, Accessories, Clothing or Others
location	String	Place where the item was lost or found
date	LocalDate	Date of the loss or discovery
description	String	Short description (colour, marks, and so on)
reporter, contact	String	Name and 10-digit phone number of the person who made the report
status	String	OPEN until the item is returned, then RETURNED
claimedBy	String	Name of the person the item was returned to (- while open)

5.2 Smart matching rule

A lost report is compared only with **open found reports** (and a found report only with open lost reports). Each pair gets a score out of 100 from four checks. Pairs that score 50 or more are shown as possible matches, best first.

Check	Points
Same category	40
Same location (ignoring case)	30
The item names share a word of at least 3 letters	20
The two dates are at most 3 days apart	10

Examples from the outputs in Section 6: the lost *Samsung Galaxy Phone* and the found *Samsung Phone* score 40 + 30 + 20 + 10 = **100%**. The lost *Student ID Card* (Canteen) and the found *ID Card* (Main Gate) score 40 + 0 + 20 + 10 = **70%** because the places differ.

5.3 Data storage

Each report is one line of **data/items.txt** with eleven fields separated by the pipe character (|):

```
id | type | name | category | location | date | description | reporter | contact | status | claimedBy
```

The file below was written by the program during the sessions in Section 6 (the two returned reports show RETURNED and the name of the owner). Long lines are wrapped here only to fit the page.

```
1|LOST|Samsung Galaxy Phone|Electronics|Library|2026-10-20|Black cover with a crack|Ananya
      Rao|9876500001|RETURNED|Ananya Rao
2|LOST|Java Programming Book|Books|Block B|2026-10-21|Blue cover, name on first page|Rahul
      Sharma|9876500002|OPEN|-
```

```
3|LOST|Student ID Card|ID Cards|Canteen|2026-10-22|CSE department card|Priya
      Nair|9876500003|OPEN|-
4|FOUND|Samsung Phone|Electronics|Library|2026-10-21|Found on a reading table|Security
      Desk|9876500010|RETURNED|Ananya Rao
5|FOUND|ID Card|ID Cards|Main Gate|2026-10-23|Card with a red lanyard|Security
      Desk|9876500010|OPEN|-
6|FOUND|Black Wallet|Accessories|Parking Area|2026-10-23|Leather wallet, no cards|Gate
      Guard|9876500011|OPEN|-
```

Output 1: Contents of data/items.txt

5.4 Program flow

Step	Action
1	Start: load all reports from data/items.txt into the ArrayList (if the file exists).
2	Show the menu and read the choice (validated to be between 0 and 7).
3	Run the selected module: report, view, search, smart match, return or statistics.
4	Reporting an item and marking an item as returned save the list to the file immediately.
5	Return to step 2 until the user enters 0, then display the thank-you message.

6. Implementation

The complete program is one file, **LostAndFoundSystem.java**. The code is shown below section by section, exactly as it appears in the file, and every operation is followed by the **real output** produced when the program was compiled and run. In the outputs, the text typed by the user appears after each prompt.

6.1 Imports and the Item Class

Item stores one report. `toFileLine()` and `fromFileLine()` convert between an object and a line of the data file; a line with the wrong number of fields or an invalid date is ignored when the file is read.

```
import java.io.*;
import java.time.LocalDate;
import java.time.format.DateTimeParseException;
import java.time.temporal.ChronoUnit;
import java.util.*;

public class LostAndFoundSystem {
    // ------------------------------------------------------------
    // Model class
    // ------------------------------------------------------------
    static class Item {
        private final int id;
        private final String type;          // LOST or FOUND
        private final String name;
        private final String category;
        private final String location;
        private final LocalDate date;
        private final String description;
        private final String reporter;
        private final String contact;
        private String status;              // OPEN or RETURNED
        private String claimedBy;           // "-" until the item is returned

        Item(int id, String type, String name, String category, String location,
             LocalDate date, String description, String reporter, String contact,
             String status, String claimedBy) {
            this.id = id;
            this.type = type;
            this.name = name;
            this.category = category;
            this.location = location;
            this.date = date;
            this.description = description;
            this.reporter = reporter;
            this.contact = contact;
            this.status = status;
            this.claimedBy = claimedBy;
        }

        int getId()             { return id; }
        String getType()        { return type; }
        String getName()        { return name; }
        String getCategory()    { return category; }
        String getLocation()    { return location; }
        LocalDate getDate()     { return date; }
        String getDescription() { return description; }
        String getReporter()    { return reporter; }
        String getContact()     { return contact; }
        String getStatus()      { return status; }
        String getClaimedBy()   { return claimedBy; }

        void setStatus(String status)       { this.status = status; }
        void setClaimedBy(String claimedBy) { this.claimedBy = claimedBy; }

        boolean isOpen() { return status.equals("OPEN"); }

        String toFileLine() {
```

```
            return String.join("|", String.valueOf(id), type, name, category,
                location, date.toString(), description, reporter, contact,
                status, claimedBy);
        }

        static Item fromFileLine(String line) {
            String[] p = line.split("\\|", -1);
            if (p.length != 11) return null;
            try {
                return new Item(Integer.parseInt(p[0]), p[1], p[2], p[3], p[4],
                        LocalDate.parse(p[5]), p[6], p[7], p[8], p[9], p[10]);
            } catch (NumberFormatException | DateTimeParseException ex) {
                return null;
            }
        }
    }
```

6.2 Data Members and the Main Menu

The reports are kept in an `ArrayList` while the program runs. The six categories are fixed so that matching can compare them exactly. The **main** method loads the file and repeats a do-while loop that shows the menu and calls the selected module until 0 is entered.

```
    // ------------------------------------------------------------
    // Data members
    // ------------------------------------------------------------
    private static final String DATA_DIR = "data";
    private static final String DATA_FILE = "data/items.txt";
    private static final String[] CATEGORIES =
        {"Electronics", "Books", "ID Cards", "Accessories", "Clothing", "Others"};
    private static final int MATCH_LIMIT = 50;
    private static final Scanner in = new Scanner(System.in);
    private static final List<Item> items = new ArrayList<>();

    // ------------------------------------------------------------
    // Program entry point
    // ------------------------------------------------------------
    public static void main(String[] args) {
        loadItems();
        int choice;

        do {
            printMenu();
            choice = readInt("Enter your choice: ", 0, 7);

            switch (choice) {
                case 1: reportItem("LOST");  break;
                case 2: reportItem("FOUND"); break;
                case 3: viewItems();         break;
                case 4: searchItems();       break;
                case 5: smartMatch();        break;
                case 6: markReturned();      break;
                case 7: showStatistics();    break;
                case 0:
                    System.out.println("Thank you for using the Smart Campus "
                        + "Lost and Found System.");
                    break;
            }
        } while (choice != 0);
    }

    private static void printMenu() {
        System.out.println();
        System.out.println("========== SMART CAMPUS LOST AND FOUND SYSTEM ==========");
        System.out.println(" 1. Report Lost Item");
        System.out.println(" 2. Report Found Item");
        System.out.println(" 3. View Items");
        System.out.println(" 4. Search Items");
        System.out.println(" 5. Smart Match for a Lost Item");
        System.out.println(" 6. Mark Item as Returned");
        System.out.println(" 7. Statistics Report");
        System.out.println(" 0. Exit");
```

```
        System.out.println("============================================================");
    }
```

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
============================================================
Enter your choice:
```

Output 2: Main menu shown when the program starts

6.3 File Handling

loadItems() reads data/items.txt at start-up. **saveItems()** creates the data folder if needed and rewrites the whole file.

```
    // ------------------------------------------------------------
    // File handling
    // ------------------------------------------------------------
    private static void loadItems() {
        File file = new File(DATA_FILE);
        if (!file.exists()) return;

        try (BufferedReader br = new BufferedReader(new FileReader(file))) {
            String line;
            while ((line = br.readLine()) != null) {
                if (line.trim().isEmpty()) continue;
                Item item = Item.fromFileLine(line);
                if (item != null) items.add(item);
            }
        } catch (IOException ex) {
            System.out.println("Could not read data file: " + ex.getMessage());
        }
    }

    private static void saveItems() {
        new File(DATA_DIR).mkdirs();
        try (PrintWriter out = new PrintWriter(new FileWriter(DATA_FILE))) {
            for (Item item : items) {
                out.println(item.toFileLine());
            }
        } catch (IOException ex) {
            System.out.println("Could not save data file: " + ex.getMessage());
        }
    }
```

6.4 Input Helpers with Validation

Every input goes through a helper that keeps asking until the value is correct: **readInt()** (a number in a range), **readText()** (not empty, no | character), **readDate()** (a real date in yyyy-MM-dd format) and **readContact()** (exactly 10 digits). **chooseCategory()** shows the numbered category list, and **nextId()** never reuses an ID.

```
    // ------------------------------------------------------------
    // Input helpers with validation
    // ------------------------------------------------------------
    private static int readInt(String prompt, int min, int max) {
        while (true) {
            System.out.print(prompt);
            String line = in.nextLine().trim();
            try {
                int value = Integer.parseInt(line);
                if (value >= min && value <= max) return value;
            } catch (NumberFormatException ignored) { }
            System.out.println("Invalid input. Enter a number between "
```

```
            + min + " and " + max + ".");
        }
    }

    private static String readText(String prompt) {
        while (true) {
            System.out.print(prompt);
            String line = in.nextLine().trim();
            if (!line.isEmpty() && !line.contains("|")) return line;
            System.out.println("Field cannot be empty or contain the | character.");
        }
    }

    private static LocalDate readDate(String prompt) {
        while (true) {
            System.out.print(prompt);
            String line = in.nextLine().trim();
            try {
                return LocalDate.parse(line);
            } catch (DateTimeParseException ex) {
                System.out.println("Invalid date. Use the format yyyy-MM-dd "
                    + "(for example 2026-10-21).");
            }
        }
    }

    private static String readContact(String prompt) {
        while (true) {
            System.out.print(prompt);
            String line = in.nextLine().trim();
            if (line.matches("\\d{10}")) return line;
            System.out.println("Enter a valid 10-digit phone number.");
        }
    }

    private static String chooseCategory() {
        System.out.println("Category:");
        for (int i = 0; i < CATEGORIES.length; i++) {
            System.out.println(" " + (i + 1) + ". " + CATEGORIES[i]);
        }
        int choice = readInt("Choose category (1-" + CATEGORIES.length + "): ",
                             1, CATEGORIES.length);
        return CATEGORIES[choice - 1];
    }

    private static Item findById(int id) {
        for (Item item : items) {
            if (item.getId() == id) return item;
        }
        return null;
    }

    private static int nextId() {
        int max = 0;
        for (Item item : items) {
            if (item.getId() > max) max = item.getId();
        }
        return max + 1;
    }
```

6.5 Table Printing

cut() shortens a long text with .. so that the columns stay aligned, and **printHeader()** and **printRow()** print the report table with `printf`.

```
    // ------------------------------------------------------------
    // Table printing
    // ------------------------------------------------------------
    private static String cut(String text, int width) {
        return text.length() <= width ? text : text.substring(0, width - 2) + "..";
    }
```

```
    private static void printLine() {
        System.out.println("----------------------------------------"
            + "-------------------------------------");
    }

    private static void printHeader() {
        printLine();
        System.out.printf("%-3s %-5s %-20s %-12s %-14s %-10s %-8s%n",
            "ID", "Type", "Item", "Category", "Location", "Date", "Status");
        printLine();
    }

    private static void printRow(Item item) {
        System.out.printf("%-3d %-5s %-20s %-12s %-14s %-10s %-8s%n",
            item.getId(), item.getType(), cut(item.getName(), 20),
            item.getCategory(), cut(item.getLocation(), 14),
            item.getDate(), item.getStatus());
    }
```

6.6 Report a Lost or Found Item

One method, **reportItem(type)**, serves menu options 1 and 2. It reads the details, creates an **Item** with status OPEN, saves it and immediately calls **showMatches()**. The first two outputs report three lost items; no found items exist yet, so there are no matches.

```
    // ------------------------------------------------------------
    // 1 and 2. Report a lost or found item
    // ------------------------------------------------------------
    private static void reportItem(String type) {
        boolean lost = type.equals("LOST");
        System.out.println();
        System.out.println("--- Report " + (lost ? "Lost" : "Found") + " Item ---");

        String name = readText("Item name: ");
        String category = chooseCategory();
        String location = readText(lost ? "Where was it lost? "
                                        : "Where was it found? ");
        LocalDate date = readDate("Date (yyyy-MM-dd): ");
        String description = readText("Short description: ");
        String reporter = readText("Your name: ");
        String contact = readContact("Contact number (10 digits): ");

        Item item = new Item(nextId(), type, name, category, location, date,
                             description, reporter, contact, "OPEN", "-");
        items.add(item);
        saveItems();

        System.out.println((lost ? "Lost" : "Found")
            + " item reported successfully! ID = " + item.getId());
        showMatches(item);
    }
```

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 1

--- Report Lost Item ---
Item name: Samsung Galaxy Phone
Category:
 1. Electronics
 2. Books
 3. ID Cards
 4. Accessories
 5. Clothing
 6. Others
```

```
Choose category (1-6): 1
Where was it lost? Library
Date (yyyy-MM-dd): 2026-10-20
Short description: Black cover with a crack
Your name: Ananya Rao
Contact number (10 digits): 9876500001
Lost item reported successfully! ID = 1

No matching reports found yet.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 3: Reporting the first lost item (ID 1) and exiting

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 1

--- Report Lost Item ---
Item name: Java Programming Book
Category:
 1. Electronics
 2. Books
 3. ID Cards
 4. Accessories
 5. Clothing
 6. Others
Choose category (1-6): 2
Where was it lost? Block B
Date (yyyy-MM-dd): 2026-10-21
Short description: Blue cover, name on first page
Your name: Rahul Sharma
Contact number (10 digits): 9876500002
Lost item reported successfully! ID = 2

No matching reports found yet.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 1

--- Report Lost Item ---
Item name: Student ID Card
Category:
 1. Electronics
 2. Books
 3. ID Cards
 4. Accessories
 5. Clothing
 6. Others
```

```
Choose category (1-6): 3
Where was it lost? Canteen
Date (yyyy-MM-dd): 2026-10-22
Short description: CSE department card
Your name: Priya Nair
Contact number (10 digits): 9876500003
Lost item reported successfully! ID = 3

No matching reports found yet.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 4: Reporting two more lost items (IDs 2 and 3) in the next run

6.7 Smart Matching

matchScore() applies the four checks of Section 5.2. **showMatches()** collects the open reports of the opposite type that score 50 or more, sorts them by score and prints them with the contact person of the best match. In the next output, three found items are reported: the phone matches the lost phone at 100%, the ID card matches the lost ID card at 70%, and the wallet matches nothing.

```
    // ------------------------------------------------------------
    // Smart matching
    // ------------------------------------------------------------
    private static boolean sharesWord(String a, String b) {
        Set<String> words = new HashSet<>();
        for (String w : a.toLowerCase().split("[^a-z0-9]+")) {
            if (w.length() >= 3) words.add(w);
        }
        for (String w : b.toLowerCase().split("[^a-z0-9]+")) {
            if (w.length() >= 3 && words.contains(w)) return true;
        }
        return false;
    }

    private static int matchScore(Item a, Item b) {
        int score = 0;
        if (a.getCategory().equals(b.getCategory()))       score += 40;
        if (a.getLocation().equalsIgnoreCase(b.getLocation())) score += 30;
        if (sharesWord(a.getName(), b.getName()))          score += 20;
        if (Math.abs(ChronoUnit.DAYS.between(a.getDate(), b.getDate())) <= 3)
            score += 10;
        return score;
    }

    private static void showMatches(Item base) {
        List<Item> matches = new ArrayList<>();
        for (Item other : items) {
            if (!other.getType().equals(base.getType()) && other.isOpen()
                    && matchScore(base, other) >= MATCH_LIMIT) {
                matches.add(other);
            }
        }

        System.out.println();
        if (matches.isEmpty()) {
            System.out.println("No matching reports found yet.");
            return;
        }

        matches.sort((x, y) -> matchScore(base, y) - matchScore(base, x));
```

```
        System.out.println("--- Possible Matches (score " + MATCH_LIMIT
            + "% or more) ---");
        printLine();
        System.out.printf("%-3s %-5s %-20s %-14s %-10s %-6s%n",
            "ID", "Type", "Item", "Location", "Date", "Match");
        printLine();
        for (Item m : matches) {
            System.out.printf("%-3d %-5s %-20s %-14s %-10s %3d%%%n",
                m.getId(), m.getType(), cut(m.getName(), 20),
                cut(m.getLocation(), 14), m.getDate(), matchScore(base, m));
        }
        System.out.println("Contact " + matches.get(0).getReporter() + " on "
            + matches.get(0).getContact() + " to verify the best match.");
    }
```

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
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 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 2

--- Report Found Item ---
Item name: Samsung Phone
Category:
 1. Electronics
 2. Books
 3. ID Cards
 4. Accessories
 5. Clothing
 6. Others
Choose category (1-6): 1
Where was it found? Library
Date (yyyy-MM-dd): 2026-10-21
Short description: Found on a reading table
Your name: Security Desk
Contact number (10 digits): 9876500010
Found item reported successfully! ID = 4

--- Possible Matches (score 50% or more) ---
-------------------------------------------------------------------------------
ID  Type  Item                Location       Date       Match
-------------------------------------------------------------------------------
1   LOST  Samsung Galaxy Phone Library        2026-10-20 100%
Contact Ananya Rao on 9876500001 to verify the best match.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
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 4. Search Items
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 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 2

--- Report Found Item ---
Item name: ID Card
Category:
 1. Electronics
 2. Books
 3. ID Cards
 4. Accessories
 5. Clothing
 6. Others
Choose category (1-6): 3
Where was it found? Main Gate
Date (yyyy-MM-dd): 2026-10-23
Short description: Card with a red lanyard
```

```
Your name: Security Desk
Contact number (10 digits): 9876500010
Found item reported successfully! ID = 5

--- Possible Matches (score 50% or more) ---
------------------------------------------------------------------------------
ID  Type  Item                Location       Date       Match
------------------------------------------------------------------------------
3   LOST  Student ID Card     Canteen        2026-10-22  70%
Contact Priya Nair on 9876500003 to verify the best match.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 2

--- Report Found Item ---
Item name: Black Wallet
Category:
 1. Electronics
 2. Books
 3. ID Cards
 4. Accessories
 5. Clothing
 6. Others
Choose category (1-6): 4
Where was it found? Parking Area
Date (yyyy-MM-dd): 2026-10-23
Short description: Leather wallet, no cards
Your name: Gate Guard
Contact number (10 digits): 9876500011
Found item reported successfully! ID = 6

No matching reports found yet.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 5: Reporting three found items – matches are shown automatically

6.8 View Items

The user chooses one of four filters. The output below uses each filter once: all items, lost only, found only and open only. The reports saved in the earlier runs are loaded from the file.

```
    // ------------------------------------------------------------
    // 3. View items
    // ------------------------------------------------------------
    private static void viewItems() {
        System.out.println();
        System.out.println("Show: 1. All  2. Lost only  3. Found only  4. Open only");
        int filter = readInt("Choose filter (1-4): ", 1, 4);

        List<Item> list = new ArrayList<>();
        for (Item item : items) {
            if (filter == 1
                    || (filter == 2 && item.getType().equals("LOST"))
                    || (filter == 3 && item.getType().equals("FOUND"))
```

```
                    || (filter == 4 && item.isOpen())) {
                list.add(item);
            }
        }

        System.out.println();
        if (list.isEmpty()) {
            System.out.println("No items found.");
            return;
        }

        System.out.println("--- Items ---");
        printHeader();
        for (Item item : list) {
            printRow(item);
        }
        System.out.println("Total items shown: " + list.size());
    }
```

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=======================================================
Enter your choice: 3

Show: 1. All  2. Lost only  3. Found only  4. Open only
Choose filter (1-4): 1

--- Items ---
-------------------------------------------------------------------------------
ID  Type  Item                Category     Location       Date       Status
-------------------------------------------------------------------------------
1   LOST  Samsung Galaxy Phone Electronics  Library        2026-10-20 OPEN
2   LOST  Java Programming B.. Books        Block B        2026-10-21 OPEN
3   LOST  Student ID Card      ID Cards     Canteen        2026-10-22 OPEN
4   FOUND Samsung Phone        Electronics  Library        2026-10-21 OPEN
5   FOUND ID Card              ID Cards     Main Gate      2026-10-23 OPEN
6   FOUND Black Wallet         Accessories  Parking Area   2026-10-23 OPEN
Total items shown: 6

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=======================================================
Enter your choice: 3

Show: 1. All  2. Lost only  3. Found only  4. Open only
Choose filter (1-4): 2

--- Items ---
-------------------------------------------------------------------------------
ID  Type  Item                Category     Location       Date       Status
-------------------------------------------------------------------------------
1   LOST  Samsung Galaxy Phone Electronics  Library        2026-10-20 OPEN
2   LOST  Java Programming B.. Books        Block B        2026-10-21 OPEN
3   LOST  Student ID Card      ID Cards     Canteen        2026-10-22 OPEN
Total items shown: 3

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
```

```
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 3

Show: 1. All  2. Lost only  3. Found only  4. Open only
Choose filter (1-4): 3

--- Items ---
-------------------------------------------------------------------------------
ID  Type  Item                Category     Location       Date       Status
-------------------------------------------------------------------------------
4   FOUND Samsung Phone       Electronics  Library        2026-10-21 OPEN
5   FOUND ID Card             ID Cards     Main Gate      2026-10-23 OPEN
6   FOUND Black Wallet        Accessories  Parking Area   2026-10-23 OPEN
Total items shown: 3

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 3

Show: 1. All  2. Lost only  3. Found only  4. Open only
Choose filter (1-4): 4

--- Items ---
-------------------------------------------------------------------------------
ID  Type  Item                Category     Location       Date       Status
-------------------------------------------------------------------------------
1   LOST  Samsung Galaxy Phone Electronics  Library        2026-10-20 OPEN
2   LOST  Java Programming B.. Books        Block B        2026-10-21 OPEN
3   LOST  Student ID Card     ID Cards     Canteen        2026-10-22 OPEN
4   FOUND Samsung Phone       Electronics  Library        2026-10-21 OPEN
5   FOUND ID Card             ID Cards     Main Gate      2026-10-23 OPEN
6   FOUND Black Wallet        Accessories  Parking Area   2026-10-23 OPEN
Total items shown: 6

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 6: Viewing items with the four filters

6.9 Search Items

The keyword is compared (ignoring case) with the ID, name, category, location and description of every report. In the output, **library** finds two reports; **card** finds three, because the wallet's description says "no cards"; and **xyz** finds nothing.

```
    // ------------------------------------------------------------
    // 4. Search items
    // ------------------------------------------------------------
    private static void searchItems() {
        System.out.println();
        System.out.print("Enter ID, item name, category, location or keyword: ");
        String key = in.nextLine().trim().toLowerCase();

        if (key.isEmpty()) {
```

```
            System.out.println("Please enter a search term.");
            return;
        }

        List<Item> found = new ArrayList<>();
        for (Item item : items) {
            if (String.valueOf(item.getId()).equals(key)
                    || item.getName().toLowerCase().contains(key)
                    || item.getCategory().toLowerCase().contains(key)
                    || item.getLocation().toLowerCase().contains(key)
                    || item.getDescription().toLowerCase().contains(key)) {
                found.add(item);
            }
        }

        if (found.isEmpty()) {
            System.out.println("No matching items found.");
            return;
        }

        System.out.println("--- Search Results ---");
        printHeader();
        for (Item item : found) {
            printRow(item);
        }
        System.out.println(found.size() + " record(s) found.");
    }
```

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 4

Enter ID, item name, category, location or keyword: library
--- Search Results ---
-------------------------------------------------------------------------------
ID  Type  Item                Category     Location       Date       Status
-------------------------------------------------------------------------------
1   LOST  Samsung Galaxy Phone Electronics  Library        2026-10-20 OPEN
4   FOUND Samsung Phone        Electronics  Library        2026-10-21 OPEN
2 record(s) found.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 4

Enter ID, item name, category, location or keyword: card
--- Search Results ---
-------------------------------------------------------------------------------
ID  Type  Item                Category     Location       Date       Status
-------------------------------------------------------------------------------
3   LOST  Student ID Card      ID Cards     Canteen        2026-10-22 OPEN
5   FOUND ID Card              ID Cards     Main Gate      2026-10-23 OPEN
6   FOUND Black Wallet         Accessories  Parking Area   2026-10-23 OPEN
3 record(s) found.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
```

```
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 4

Enter ID, item name, category, location or keyword: xyz
No matching items found.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 7: Searching by location, by keyword and with no match

6.10 Smart Match for a Lost Item

Option 5 runs the matching on demand for a lost report. The output shows a lost ID card matched with the found ID card (70%), a lost book with no match, a found report (rejected, because only lost items can be matched) and an unknown ID.

```
    // ------------------------------------------------------------
    // 5. Smart match for a lost item
    // ------------------------------------------------------------
    private static void smartMatch() {
        System.out.println();
        int id = readInt("Enter ID of the lost item: ", 1, Integer.MAX_VALUE);
        Item item = findById(id);

        if (item == null) {
            System.out.println("Item with ID " + id + " not found.");
            return;
        }
        if (!item.getType().equals("LOST")) {
            System.out.println("Item " + id + " is a FOUND report. "
                + "Smart match works for LOST items.");
            return;
        }
        if (!item.isOpen()) {
            System.out.println("Item " + id + " is already marked as "
                + item.getStatus() + ".");
            return;
        }

        System.out.println("Searching matches for: " + item.getName()
            + " (" + item.getCategory() + ", " + item.getLocation() + ", "
            + item.getDate() + ")");
        showMatches(item);
    }
```

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 5

Enter ID of the lost item: 3
Searching matches for: Student ID Card (ID Cards, Canteen, 2026-10-22)
```

```
--- Possible Matches (score 50% or more) ---
--------------------------------------------------------------------------------
ID  Type  Item                Location       Date       Match
--------------------------------------------------------------------------------
5   FOUND ID Card             Main Gate      2026-10-23  70%
Contact Security Desk on 9876500010 to verify the best match.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 5

Enter ID of the lost item: 2
Searching matches for: Java Programming Book (Books, Block B, 2026-10-21)

No matching reports found yet.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 5

Enter ID of the lost item: 4
Item 4 is a FOUND report. Smart match works for LOST items.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 5

Enter ID of the lost item: 99
Item with ID 99 not found.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 8: Smart match for lost items, with two invalid requests

6.11 Mark Item as Returned

After the owner is verified, the item is marked RETURNED together with the owner's name, after a y/n confirmation. A related report of the opposite type can be closed in the same step. In the output, found item 4 is returned to Ananya Rao and the related lost report 1 is closed too; then the same item is tried again, an unknown ID is tried, and one return is cancelled with n. The second output shows that only four reports remain open.

```
    // ------------------------------------------------------------
    // 6. Mark item as returned
    // ------------------------------------------------------------
    private static void markReturned() {
        System.out.println();
        int id = readInt("Enter ID of the item to mark as returned: ",
                         1, Integer.MAX_VALUE);
        Item item = findById(id);

        if (item == null) {
            System.out.println("Item with ID " + id + " not found.");
            return;
        }
        if (!item.isOpen()) {
            System.out.println("Item " + id + " is already marked as "
                + item.getStatus() + ".");
            return;
        }

        String owner = readText("Returned to / claimed by (name): ");
        System.out.print("Mark \"" + item.getName() + "\" as returned to "
            + owner + "? (y/n): ");
        String answer = in.nextLine().trim().toLowerCase();

        if (!answer.equals("y")) {
            System.out.println("Action cancelled.");
            return;
        }

        item.setStatus("RETURNED");
        item.setClaimedBy(owner);

        int related = readInt("Related report ID to close as well (0 for none): ",
                              0, Integer.MAX_VALUE);
        if (related != 0) {
            Item other = findById(related);
            if (other == null || !other.isOpen()
                    || other.getType().equals(item.getType())) {
                System.out.println("Report " + related + " cannot be closed "
                    + "with this item.");
            } else {
                other.setStatus("RETURNED");
                other.setClaimedBy(owner);
                System.out.println("Related report " + related + " closed too.");
            }
        }

        saveItems();
        System.out.println("Item marked as RETURNED.");
    }
```

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
============================================================
Enter your choice: 6

Enter ID of the item to mark as returned: 4
```

```
Returned to / claimed by (name): Ananya Rao
Mark "Samsung Phone" as returned to Ananya Rao? (y/n): y
Related report ID to close as well (0 for none): 1
Related report 1 closed too.
Item marked as RETURNED.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 6

Enter ID of the item to mark as returned: 4
Item 4 is already marked as RETURNED.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 6

Enter ID of the item to mark as returned: 99
Item with ID 99 not found.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 6

Enter ID of the item to mark as returned: 5
Returned to / claimed by (name): Priya Nair
Mark "ID Card" as returned to Priya Nair? (y/n): n
Action cancelled.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 9: Returning an item, closing the related report, and handling invalid cases

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
```

```
=========================================================
Enter your choice: 3

Show: 1. All  2. Lost only  3. Found only  4. Open only
Choose filter (1-4): 4

--- Items ---
-------------------------------------------------------------------------------
ID  Type  Item                Category     Location       Date       Status
-------------------------------------------------------------------------------
2   LOST  Java Programming B.. Books        Block B        2026-10-21 OPEN
3   LOST  Student ID Card      ID Cards     Canteen        2026-10-22 OPEN
5   FOUND ID Card              ID Cards     Main Gate      2026-10-23 OPEN
6   FOUND Black Wallet         Accessories  Parking Area   2026-10-23 OPEN
Total items shown: 4

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 10: Open items after the return (reports 1 and 4 are no longer listed)

6.12 Statistics Report

The report counts lost and found reports, open and returned reports, the recovery rate (returned reports as a percentage of all reports), the reports per category and the location with the most reports. The closing brace at the end of this listing ends the class.

```
    // ------------------------------------------------------------
    // 7. Statistics report
    // ------------------------------------------------------------
    private static void showStatistics() {
        System.out.println();
        if (items.isEmpty()) {
            System.out.println("No reports found.");
            return;
        }

        int lost = 0, found = 0, open = 0, returned = 0;
        Map<String, Integer> categories = new TreeMap<>();
        Map<String, Integer> places = new TreeMap<>();

        for (Item item : items) {
            if (item.getType().equals("LOST")) lost++; else found++;
            if (item.isOpen()) open++; else returned++;
            categories.merge(item.getCategory(), 1, Integer::sum);
            places.merge(item.getLocation(), 1, Integer::sum);
        }

        String busiest = null;
        for (Map.Entry<String, Integer> e : places.entrySet()) {
            if (busiest == null || e.getValue() > places.get(busiest)) {
                busiest = e.getKey();
            }
        }

        System.out.println("--- Statistics Report ---");
        System.out.println("Total reports    : " + items.size());
        System.out.println("Lost reports     : " + lost);
        System.out.println("Found reports    : " + found);
        System.out.println("Open             : " + open);
        System.out.println("Returned         : " + returned);
        System.out.printf("Recovery rate    : %.1f%%%n",
            returned * 100.0 / items.size());
```

```
        System.out.println("By category      : " + categories);
        System.out.println("Busiest location : " + busiest
            + " (" + places.get(busiest) + " report(s))");
    }
}
```

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=======================================================
Enter your choice: 7

--- Statistics Report ---
Total reports    : 6
Lost reports     : 3
Found reports    : 3
Open             : 4
Returned         : 2
Recovery rate    : 33.3%
By category      : {Accessories=1, Books=1, Electronics=2, ID Cards=2}
Busiest location : Library (2 report(s))

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=======================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 11: Statistics report (2 of 6 reports returned = 33.3%)

6.13 Input Validation

The last run enters wrong values on purpose: a menu choice of 9 or *abc*, an empty item name, a name containing |, category 7, the impossible date 2026-13-45, the text *tomorrow*, and phone numbers *12345* and *98765x*. Each one is rejected and the question is asked again. The new report gets ID 7.

```
========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=======================================================
Enter your choice: 9
Invalid input. Enter a number between 0 and 7.
Enter your choice: abc
Invalid input. Enter a number between 0 and 7.
Enter your choice: 1

--- Report Lost Item ---
Item name:
Field cannot be empty or contain the | character.
Item name: A|B
Field cannot be empty or contain the | character.
Item name: Umbrella
Category:
 1. Electronics
 2. Books
 3. ID Cards
```

```
 4. Accessories
 5. Clothing
 6. Others
Choose category (1-6): 7
Invalid input. Enter a number between 1 and 6.
Choose category (1-6): 6
Where was it lost? Hostel
Date (yyyy-MM-dd): 2026-13-45
Invalid date. Use the format yyyy-MM-dd (for example 2026-10-21).
Date (yyyy-MM-dd): tomorrow
Invalid date. Use the format yyyy-MM-dd (for example 2026-10-21).
Date (yyyy-MM-dd): 2026-10-24
Short description: Green umbrella
Your name: Kiran Kumar
Contact number (10 digits): 12345
Enter a valid 10-digit phone number.
Contact number (10 digits): 98765x
Enter a valid 10-digit phone number.
Contact number (10 digits): 9876500004
Lost item reported successfully! ID = 7

No matching reports found yet.

========== SMART CAMPUS LOST AND FOUND SYSTEM ==========
 1. Report Lost Item
 2. Report Found Item
 3. View Items
 4. Search Items
 5. Smart Match for a Lost Item
 6. Mark Item as Returned
 7. Statistics Report
 0. Exit
=========================================================
Enter your choice: 0
Thank you for using the Smart Campus Lost and Found System.
```

Output 12: Validation of wrong inputs while reporting a lost item

7. Test Results

Test	Input	Result shown by the program	Result
Report lost item	Samsung Galaxy Phone, Library	Lost item reported successfully! ID = 1	Pass
No match yet	First lost reports	No matching reports found yet.	Pass
Auto match, strong	Found: Samsung Phone, Library	Matches lost ID 1 with 100%	Pass
Auto match, partial	Found: ID Card, Main Gate	Matches lost ID 3 with 70%	Pass
No match	Found: Black Wallet	No matching reports found yet.	Pass
View filters	All / Lost / Found / Open	6 / 3 / 3 / 6 items listed	Pass
Search	library, card, xyz	2, 3 and 0 record(s)	Pass
Smart match request	Lost ID 3; found ID 4; ID 99	Match 70%; rejected; not found	Pass
Return item	ID 4, Ananya Rao, related 1	Both reports marked RETURNED	Pass
Return again / cancel	ID 4; ID 5 with n	Already returned; action cancelled	Pass
Statistics	Option 7	Recovery rate 33.3%, busiest location Library	Pass
Validation	Bad choice, date, phone, category	Error message and input asked again	Pass
Persistence	Restart the program	Reports loaded from data/items.txt	Pass

8. How to Run

Compile and run the program from the folder that contains the source file:

```
javac LostAndFoundSystem.java
java LostAndFoundSystem
```

The program creates a **data** folder next to it, containing items.txt, the first time a report is saved.

9. Conclusion and Future Scope

The Smart Campus Lost and Found System replaces scattered notices with one searchable record and speeds up recovery by suggesting matches automatically. It uses core Java only: a model class, collections, date handling, file handling and validated console input.

- Send an SMS or email to the owner as soon as a match is found.
- Attach a photograph of the item to each report.
- A Swing or web interface for phones, with a database (MySQL / SQLite) instead of a text file.
- Login for security staff, and a verification question before an item is handed over.
- A better matching rule that also compares the words in the description.